

# Zac Shi

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## EDUCATION

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### University of California, Los Angeles

B.S. in Mathematics

Minor in Data Science Engineering

GPA: 3.93

*Los Angeles, CA*

*Sep 2024 – Present*

## RESEARCH & PROJECTS

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### Varney Geospace Research Group

*Undergraduate Researcher*

University of California, Los Angeles

*Los Angeles, CA*

*Sep 2025 – Present*

- Analyzed 2010–2025 ionospheric radar datasets to investigate ionospheric parameter couplings through building multi-parameter physical models by defining a weighted least-squares cost function with uncertainty-aware residuals.
- Implemented Levenberg–Marquardt optimization, orthogonal distance regressions through SciPy workflow.
- Packaged the analysis as a reproducible Python pipeline (NumPy/SciPy/Matplotlib, Jupyter) for remote execution on UCLA’s computing environment.

### Sustainable Los Angeles Grand Challenge

*Research Scholar*

University of California, Los Angeles

*Los Angeles, CA*

*Sep 2025 – Present*

## EXPERIENCE

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### United Spirit Association

*Drum Major Head Instructor*

*California*

*Jun 2025 – Jul 2026*

- Taught leadership, mace flourishing, and drill to over 150 student drum majors at camps across California.

### UCLA CalTeach

*Student Teacher*

University of California, Los Angeles

*Los Angeles, CA*

*Jan 2025 – Mar 2026*

- Assisted host teacher in mathematics instruction 3 hours per week; enrolled in concurrent teaching seminar on mathematics pedagogy.

### Peacocks American Drum Major

*Parade Adjudicator*

*Los Angeles, CA*

*May 2025 – Present*

## LEADERSHIP & ORGANIZATIONAL ROLES

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### Undergraduate Mathematics Students Association

*President*

University of California, Los Angeles

*Los Angeles, CA*

*Mar 2025 – Present*

- President of UCLA’s largest mathematics organization (500+ members); manage club budget and lead a team of officers across academics, industry, and outreach.
- Launched a sponsorship program securing partnerships with Tower Capital and QRT Trading; host 20+

events annually including research talks, networking nights, and the Integration Bee.

**UCLA Solid Gold Sound Marching Band**

*Trombone Section Leader*

*Los Angeles, CA*

*Sep 2026*

TECHNICAL SKILLS

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- **Languages:** Python, C++, R
- **Typeset:** L<sup>A</sup>T<sub>E</sub>X
- **Software:** DaVinci Resolve, Adobe Photoshop, Adobe Lightroom